

$\sigma_{ab}(i)$ — the shielding tensor
what NMR reads at the nucleus

adds a field

perturbs the
response

MAGNETIC *induced moments*

field $B_a = D_{ab} m_b$

ring current (χ): McConnell · JB ·
HM; bond anisotropy ($\Delta\chi$)

$\cdot \mathbf{m}$

ELECTRICAL *charges*

field gradient $\partial_a E_b = D_{ab} q$

charge / EFG: Coulomb · APBS ·
MOPAC · EEQ; π quadrupole

$\cdot \mathbf{q}$

$$D_{ab} = \partial_a \partial_b \frac{1}{\rho} = \frac{3\hat{\rho}_a \hat{\rho}_b - \delta_{ab}}{\rho^3}$$

the one geometry — carries
($3 \cos^2 \theta - 1$)/ ρ^3 and the 54.7° node

$\uparrow \partial_a \partial_b$

*electric & mag-
netic become one
geometry here*

$$\frac{1}{\rho} \text{ — the Laplacian's Green's function, } \nabla^2 \frac{1}{\rho} = -4\pi\delta(\rho)$$

both the electric potential φ and the
magnetic potential $\mathbf{A}$ are built on it

static / near-field limit

Ampère / Biot–Savart

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

the current B_0 stirs up

Gauss / Coulomb

$$\nabla \cdot \mathbf{E} = \rho/\epsilon_0$$

Maxwell's two source equations — magnetic (Ampère) and electric (Gauss)